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Dry matter content as a key indicator of physiological and quality traits in sugar beet

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Abstract

Dry matter percentage (DM) is a critical physiological trait in sugar beet, functioning not only as a quantitative measure but also as an integrative marker of carbon assimilation, assimilate partitioning, and ionic regulation. We hypothesized that variation in DM across genotypes arises from differences in photosynthetic efficiency, sucrose allocation to storage roots, and mineral ion homeostasis. To test this, we evaluated 66 sugar beet cultivars across 47 field trials (2019–2024) and assessed the relationships between DM and key agronomic, quality, and physicochemical traits. DM ranged from 15.8 to 27.1%, reflecting substantial genotypic diversity together with interannual environmental variation. Spearman's rank correlation analysis revealed strong positive associations between DM and sucrose content ($r=0.88$), white sugar content ($r=0.87$), and extraction coefficient of sugar ($r=0.78$), whereas moderate positive correlations were observed with sugar yield ($r=0.34$) and white sugar yield ($r=0.54$). Conversely, DM was negatively correlated with root yield ($r=-0.22$), sodium ($r=-0.75$), nitrogen ($r=-0.31$), and molasses sugar ($r=-0.71$). Multivariate regression confirmed that DM significantly improves quality-related traits while reducing impurity-related traits, with no significant influence on potassium or alkalinity. Year-specific correlation analyses further demonstrated that the positive associations of dry matter with sugar content, white sugar content, and extractable sugar coefficient, together with its negative association with sodium concentration, remained consistent across growing seasons, whereas relationships with yield-related traits were more variable. These findings support the role of DM as a mechanistic marker of source-sink dynamics and ion regulation in sugar beet physiology. By linking DM to underlying physiological processes, this study provides a framework for selecting genotypes with enhanced sucrose accumulation and extractable sugar quality. DM thus emerges as a reliable indicator for breeding programs aiming to improve sugar quality and processing efficiency while maintaining acceptable field performance.

Keywords Association, Dry matter, Sugar beet

1 Introduction

Sugar beet (*Beta vulgaris* L.) is a fundamental industrial crop whose economic value depends on both root biomass and extractable sucrose. Among root quality traits, dry matter percentage is of particular importance because it reflects the non-water fraction



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of biomass, largely composed of sugars, and strongly influences sugar yield and processing quality. Dry matter content integrates physiological processes including photosynthetic carbon gain, assimilate partitioning into storage roots, and regulation of mineral ion uptake. While correlations between dry matter and sucrose concentration have long been reported, the physiological mechanisms underlying these associations remain poorly understood [1].

Carbon assimilation and source–sink dynamics play a central role in dry matter formation. High- dry matter genotypes may exhibit greater activity of sucrose-phosphate synthase, invertases, or sucrose synthase, which enhance sucrose loading and storage in root vacuoles [2]. Similarly, efficient phloem transport and assimilate unloading to root parenchyma promote dry matter accumulation. Ion balance is also critical: excess Na^+ and NO_3^- not only reduce sucrose purity but also alter osmotic status, thereby diluting dry matter content [3]. Thus, variation in dry matter across genotypes may reflect fundamental physiological differences in carbon metabolism and ionic regulation.

The accumulation of dry matter and sucrose is also influenced by environmental and agronomic factors such as growth duration, climatic conditions, irrigation, and nutrient management. Drought and salinity stress, for instance, can increase dry matter by reducing water content, but simultaneously decrease juice volume and processing quality. Adaptations such as increased root-to-shoot ratios under drought [4–6] highlight the dual role of dry matter in stress tolerance and product quality. Improved irrigation and balanced nitrogen application significantly enhance dry matter, sucrose content, and yield [7]. Long growth duration and optimal planting density further promote dry matter and sucrose accumulation, while continuous monoculture negatively impacts photosynthetic capacity and dry matter partitioning [8].

Genotypic variation also plays a major role. Although total dry matter production may be similar among cultivars, differences in allocation between root and shoot tissues account for much of the observed variation [9]. Breeding strategies aimed at improving root-to-shoot partitioning or enhancing sink strength are therefore promising approaches for increasing dry matter and sugar yield. Recent genomic studies have confirmed extensive genetic diversity underlying dry matter and sucrose-related traits, with complex marker-trait associations emphasizing the multifactorial control of these traits [10].

Despite extensive breeding for yield and sucrose concentration, relatively few studies have addressed dry matter as an integrative physiological trait. Most reports have remained descriptive, focusing on correlations rather than mechanisms. We hypothesize that variation in dry matter reflects differences in carbon partitioning efficiency and ion regulation, and that this integrative trait can predict sugar yield and purity. The objectives of this study were therefore to: (i) quantify variation in dry matter across cultivars and years, (ii) assess associations between dry matter and agronomic/quality traits, and (iii) interpret these associations within a physiological framework to identify mechanistic drivers of sucrose accumulation and root quality.

2 Materials and methods

This study utilized data obtained from 47 field trials conducted between 2019 and 2024 at the Motahari Research Station in Karaj, Iran. The trials involved 66 registered mono-germ diploid sugar beet cultivars (Supplementary Table S1). The physicochemical properties of the experimental soil during multi-year field trials are presented in Table 1.

Monthly meteorological data, including relative humidity, maximum temperature, minimum temperature, and temperature range are provided in Supplementary Table S2.

In autumn, agricultural operations including plowing, fertilizer application, and supplementary tillage for seedbed preparation (shallow plowing, disking, and harrowing) were performed. Seeds were sown each year in a randomized complete block design with four replications in late April. Each experimental plot consisted of three rows, with intra-row spacing of 20 cm and inter-row spacing of 50 cm. Agronomic practices, including fertilizer application, irrigation, planting depth, and crop management, were kept as uniform as possible across years to minimize management-related variation. Pest control involved chemical pesticides, while weed management was conducted manually and mechanically. During the harvest stage, roots from each plot were harvested, weighed, and homogenized into pulp samples. These samples were then transferred to specialized trays and stored at −20 °C for one day to one week before being sent to the Sugar Technology Laboratory at the Sugar Beet Seed Institute for qualitative analysis. The qualitative evaluation of pulp samples was performed using an automated beet laboratory system (Betalyser, Anton Paar, Germany), adhering to ICUMSA standard protocols [11].

Root yield and white sugar yield traits were measured. White sugar yield was calculated using Eq. (1) [12]:

$$WSY = RY \times WSC \tag{1}$$

Where WSY represents white sugar yield, RY denotes root yield, and WSC signifies the white sugar content. To evaluate the effect of dry matter on multiple agronomic and quality traits including root yield, sugar yield, white sugar yield, sugar content (SC), sodium (Na), potassium (K), alkalinity (Alc), white sugar content, extraction coefficient of sugar (ECS), and molasses sugar (Ms), separate linear mixed-effects models were fitted. The general model was:

$$Y_i = \beta_0 + \beta_1 DM_i + b_{Cultivar(i)} + \epsilon_i.$$

where, Y_i represents the response variable, β_0 is the overall intercept, β_1 represents the fixed effect of dry matter, $b_{Cultivar}$ is the random intercept associated with cultivar, assumed to follow a normal distribution $N(0, \sigma_b^2)$, and ϵ_i is the residual error, assumed

Table 1 Physical and chemical properties of soil at the experimental site of Motahari Research Satation during multi-year assessment

Year	pH	EC (ds.m ⁻¹)	O.C. (%)	P (ppm)	K (ppm)	Clay (%)	Silt (%)	Sand 9%)	Texture
2019	8.3	1.04	0.9	6.3	469	31.4	39.1	29.4	Clay-loam
2020	7.5	1.3	2.1	13.4	599	40.3	40.2	19.0	
2021	7.2	0.5	0.7	8.2	596	37.3	41.4	21.3	
2022	7.3	0.7	1.01	1.97	398	37.3	41.4	21.3	
2023	7.2	0.8	1.3	2.7	507	37.3	41.4	21.3	
2024	6.7	1.3	0.8	8.9	349	34.6	38.9	26.5	

to follow $N(0, \sigma^2)$. Including cultivar as a random effect accounted for genetic variability among cultivars while estimating the overall effect of dry matter across the combined multi-year dataset.

Models were fitted using the Maximum Likelihood (ML) method in PROC MIXED of SAS version 9.4 (SAS Institute, Cary, NC, USA). The significance of the fixed effect of dry matter was evaluated using Type III tests of fixed effects, with statistical significance declared at $P < 0.05$. Model adequacy was assessed using Akaike's Information Criterion (AIC), corrected AIC (AICC), Bayesian Information Criterion (BIC), and graphical examination of residuals, including residual-versus-fitted plots, normal Q–Q plots, and histograms with fitted normal density curves.

Spearman's rank correlation coefficients were calculated using the pooled multi-year dataset to quantify associations between dry matter and agronomic and technological traits. In addition, Pearson correlation coefficients were calculated separately for each growing season to evaluate the consistency of trait relationships under different annual environmental conditions. Correlation heatmaps were generated in R (version 4.5.1). Monthly meteorological summaries used to characterize seasonal environmental variation are presented in Supplementary Table S2, and the year-specific Pearson correlation matrices are provided in Supplementary Table S3.

All statistical analyses were conducted using SAS software version 9.4 (SAS Institute, Cary, NC, USA). Additional analyses and visualizations were performed in R to explore relationships among variables. Residual diagnostics including marginal and conditional residual plots, Q–Q plots, and histograms with overlaid normal density curves were used to verify assumptions of normality and homoscedasticity for each trait [13].

3 Results

The analysis of dry matter percentage across a diverse set of sugar beet genotypes revealed a considerable range, varying from 15.8 to 27.1%. This wide variation reflects substantial genetic diversity as well as the sensitivity of dry matter formation to environmental and management factors. The mean dry matter percentage was approximately 21.7%, consistent with earlier reports, and highlights the pivotal role of dry matter in determining sugar yield, processing quality, and tolerance to environmental stresses. Such variability presents an opportunity for targeted breeding programs aimed at selecting superior genotypes with enhanced dry matter traits.

Meteorological conditions varied among years (Supplementary Table S2). Annual precipitation ranged from 0.94 mm day⁻¹ in 2022 to 1.75 mm day⁻¹ in 2019, while maximum summer temperatures consistently exceeded 36 °C. Relative humidity showed considerable seasonal fluctuations, declining during summer months and increasing during autumn and winter.

Root yield also showed large variation, ranging from 49.7 to 140.8 t ha⁻¹, with an average of 91.5 t ha⁻¹, comparable to commercial production levels. The mean sugar content was 15.2%, while white sugar content averaged 13.4%. Mineral nutrients were highly variable: sodium (Na) ranged from 0.7 to 11.9 meq.100 g⁻¹, potassium (K) from 2.2 to 7.2 meq.100 g⁻¹, and nitrogen (N) from 0.3 to 7.1 meq.100 g⁻¹. This variability likely reflects differences in soil characteristics, environmental conditions, and fertilization practices. Importantly, such fluctuations directly affect plant physiology, as excess Na⁺ may induce

salinity stress, elevated N may alter assimilate partitioning toward amino compounds, and variable K levels influence osmotic adjustment and sugar translocation.

3.1 Correlation analysis

Spearman’s rank correlation revealed significant associations between dry matter and key agronomic and quality traits at the 1% probability level (Table 2). A weak negative correlation was observed between dry matter and root yield ($r = -0.22$), suggesting a tendency for increased dry matter to coincide with reduced fresh biomass; a trade-off between yield and quality frequently reported in sugar beet. Conversely, dry matter exhibited significant positive correlations with sugar yield ($r=0.34$), white sugar yield ($r=0.54$), sucrose content ($r=0.88$), and white sugar content ($r=0.87$). These strong coefficients emphasize that genotypes with higher dry matter not only possess denser roots but also show superior sucrose accumulation and recoverable sugar.

The extraction coefficient of sugar also correlated strongly with dry matter ($r=0.78$), supporting the hypothesis that higher dry matter enhances the efficiency of sucrose recovery during processing. Mineral composition showed contrasting patterns: Na ($r = -0.75$) and N ($r = -0.31$) were negatively associated with dry matter, implying that higher- dry matter roots maintain tighter control over ion uptake or compartmentalization, leading to fewer impurities and improved quality. Potassium ($r = -0.05$) and alkalinity ($r = -0.09$) showed weak, nonsignificant correlations, suggesting limited influence on dry matter variability. Molasses sugar content displayed a strong negative correlation with dry matter ($r = -0.71$), indicating that higher dry matter reduces by-products that hinder sugar crystallization. To evaluate whether these relationships were stable under contrasting environmental conditions, Pearson correlation coefficients between dry matter and the measured traits were calculated separately for each experimental year (Supplementary Table S3). Despite considerable interannual variation in weather conditions (Supplementary Table S2), the positive associations of dry matter with sugar content, white sugar content, and extraction coefficient of sugar remained consistently strong across all six years. Likewise, dry matter maintained a stable negative relationship with sodium concentration and, in most years, with molasses sugar content. In contrast, correlations between dry matter and root yield, sugar yield, and white sugar yield varied substantially among years, indicating a stronger environmental influence on these productivity-related traits than on the technological quality attributes.

The Pearson correlation heatmap revealed clear clusters of strongly associated traits in sugar beet roots. A pronounced positive correlation was detected among SC, WSC, and ECS ($r>0.9$), indicating that these quality-related traits are tightly interlinked (Fig. 1). Similarly, Sy and Wsy were highly correlated with each other ($r=0.95$). These strong positive linkages suggest that improvement in one of these traits is likely to be accompanied

Table 2 Spearman’s rank correlation coefficients between dry matter percentage, quantitative and qualitative traits of sugar beet cultivars in multi-year field assessment

	Ry	Sy	Wsy	Sc	Wsc	ECS	Na	K	N	Alc	Ms
	(t ha ⁻¹)			(%)			(mmol 100 g ⁻¹ beet)				(%)
DM (%)	-0.22	0.34**	0.54**	0.88**	0.87**	0.78**	-0.75**	-0.05	-0.31**	-0.09	-0.71**

*,**Significant at 5% and 1% probability level, respectively
DM: dry matter percentage; Ry: root yield; Sy: sugar yield; Wsy: white sugar yield; Sc: sugar content; Wsc: white sugar content; ECS: extraction coefficient of sugar; Na: sodium; K: potassium; N: nitrogen; Alc: alkalinity; Ms: molasses sugar

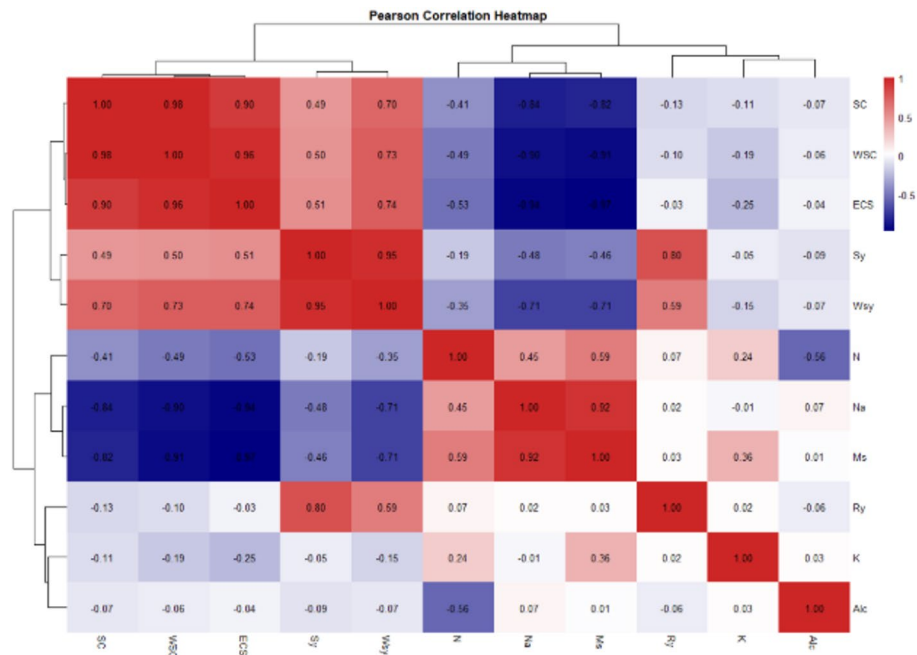


Fig. 1 Correlation heatmap of physiological and technological traits in sugar beet roots in multi-year field assessment. Ry: root yield; Sy: sugar yield; Wsy: white sugar yield; Sc: sugar content; Wsc: white sugar content; ECS: extraction coefficient of sugar; Na: sodium; K: potassium; N: nitrogen; Alc: alkalinity; Ms: molasses sugar

by parallel gains in the others, making them promising candidates for simultaneous selection in breeding programs.

In contrast, several strong negative correlations highlight potential trade-offs. Sugar yield and Wsy showed moderate negative relationships with Na and Ms, ECS showed strong negative associations with Na, K, N, and Ms ($P<0.01$), suggesting that improvements in extractable sugar percentage may occur at the expense of certain compositional or physiological attributes. Likewise, SC and WSC were negatively correlated with these same traits, reinforcing the antagonistic relationship between sugar yield components and extraction coefficient of sugar (Fig. 1). Some traits, such as Ry, K, and Alc, displayed generally weak or negligible correlations with most parameters, indicating that they can be modified more independently. Overall, the heatmap underscores the coexistence of strong synergistic relationships and marked trade-offs among agronomic and quality traits, providing valuable guidance for sugar beet breeding. Year-specific correlation analyses further showed that the main associations identified in the pooled dataset were largely consistent across years, particularly the positive relationships of dry matter with sugar content, white sugar content, and extraction coefficient of sugar, as well as its negative association with sodium concentration. In contrast, correlations between dry matter and yield-related traits exhibited greater interannual variability, suggesting a stronger influence of environmental conditions on these relationships (Supplementary Table S3).

3.2 Regression analysis

Complementing the correlation analysis, multivariate regression was performed to evaluate the independent effects of measured traits on DM percentage and to develop predictive models for DM. The mixed-effects regression model revealed that DM exerted a significant negative influence on root yield and sodium content, confirming

Table 3 Estimated fixed effects of DM on measured plant traits in multi-year field assessment using a mixed-effects model

Trait	β	SE	t	P
Ry	-1.5040	0.3445	-4.37	< 0.0001
Sy	0.3984	0.0581	6.86	< 0.0001
Wsy	0.6848	0.0563	12.17	< 0.0001
SC	0.7367	0.0225	32.68	< 0.0001
Na	-0.7332	0.0380	-19.31	< 0.0001
K	-0.0269	0.0213	-1.26	0.207
N	-0.1392	0.0244	-5.70	< 0.0001
Alc	-0.1177	0.0994	-1.18	0.237
WSC	1.0113	0.0342	29.61	< 0.0001
ECS	2.9540	0.1402	21.06	< 0.0001
Ms	-0.2735	0.0159	-17.25	< 0.0001

Ry: root yield (t ha^{-1}), Sy: sugar yield (t ha^{-1}), Wsy (t ha^{-1}), white sugar yield (t ha^{-1}), Sc: sugar content (%), Wsc: white sugar content (%), ECS: extraction coefficient of sugar (%), Na: sodium ($\text{mmol } 100 \text{ g}^{-1}$ beet), K: potassium ($\text{mmol } 100 \text{ g}^{-1}$ beet), N: nitrogen ($\text{mmol } 100 \text{ g}^{-1}$ beet), Alc: alkalinity, Ms: molasses sugar (%)

that increases in DM are associated with reductions in these variables. In contrast, DM had a highly significant positive effect on sugar yield, white sugar yield, sucrose content, white sugar content, and extraction coefficient of sugar, indicating that these quality and efficiency traits improve with higher DM values. Additionally, molasses sugar content was negatively and significantly affected by DM, while potassium and alkalinity showed nonsignificant effects, highlighting a lower sensitivity of these traits to variations in DM (Table 3).

Together, these results confirm that dry matter is not merely a descriptive trait but an integrative marker of physiological efficiency, capturing both carbon allocation and ion regulation in sugar beet roots.

4 Discussion

Our study investigates the physiological basis of dry matter variation in sugar beet by integrating large-scale phenotypic data with agronomic and quality traits. We hypothesize that dry matter is not merely a descriptive agronomic parameter but a mechanistic integrator of carbon assimilation, assimilate partitioning, and ion regulation, governed by complex physiological and molecular processes. Although all experiments were conducted at the same location, interannual variation in rainfall, air temperature, and relative humidity was evident. Such seasonal differences may partly explain the observed variability among years and should be considered when interpreting the relationships between dry matter and quality traits. Nevertheless, because the objective of the present study was to investigate overall associations across a large multi-year dataset rather than to quantify genotype $\times$ environment interaction, the analyses were performed using pooled data.

The observed strong positive correlations between dry matter and sucrose content, white sugar yield, and extraction coefficient of sugar suggest that high-dry matter genotypes allocate a greater proportion of photosynthates toward carbohydrate storage in roots. This allocation likely involves enhanced phloem loading and unloading mechanisms, mediated by sucrose transporter proteins, which regulate source-to-sink carbon flux [14–15]. Such genotypes may exhibit increased activity or expression of these transporters, facilitating sucrose accumulation and contributing to improved sugar yield and quality.

Conversely, the significant negative correlations between dry matter and sodium and nitrogen contents point toward active ion homeostasis mechanisms in high-dry matter genotypes. Efficient exclusion or compartmentalization of Na^+ through ion transporters family members may reduce ion toxicity and maintain osmotic balance [16–17]. Similarly, lower nitrogen content suggests a metabolic shift favoring carbon allocation to sucrose synthesis over amino acid biosynthesis, reflecting alterations in carbon-nitrogen metabolism that optimize sugar accumulation. These physiological adjustments likely confer enhanced tolerance to environmental stresses and improved extract purity.

While these correlations align with prior agronomic observations, our study advances the field by providing a comprehensive, multi-environment dataset supporting dry matter as a robust integrative trait linking carbon and ion dynamics in sugar beet roots. This insight opens new avenues for molecular and biochemical investigations, such as transcriptomic profiling of sucrose transporter and ion channel genes, enzyme activity assays related to carbon metabolism, and ionomic analyses to unravel genotype-specific regulatory networks.

The year-specific correlation analysis further supported the robustness of the overall relationships observed in the pooled dataset (Supplementary Table S3). Despite the interannual differences in temperature, and relative humidity (Supplementary Table S2), the positive associations of dry matter with sugar content, white sugar content, and extraction coefficient of sugar remained consistently strong across all growing seasons. Likewise, the negative relationship between dry matter and sodium concentration was maintained throughout the study period. By contrast, the relationships between dry matter and yield-related traits (root yield, sugar yield, and white sugar yield) showed greater year-to-year variation, indicating that these traits are more sensitive to seasonal environmental conditions than technological quality traits. These findings suggest that dry matter is more closely linked to the physiological processes governing sugar accumulation and impurity regulation than to biomass production per se.

Furthermore, understanding the genetic architecture underlying dry matter variation will be critical for marker-assisted breeding. Recent genomic studies [10] indicate substantial genetic diversity associated with dry matter and sucrose traits, suggesting that targeted selection for transporter and ion regulation genes could enhance breeding efficiency.

In conclusion, dry matter content emerges as a physiologically meaningful marker reflecting the interplay between carbon assimilation and ion regulation, rather than a simple correlative trait. Incorporating mechanistic insights into dry matter variation will facilitate development of superior sugar beet cultivars with optimized yield, quality, and stress resilience.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s44372-026-00865-w>.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

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Author contributions

PF and BB conceived the idea of the paper and drafted the initial manuscript; VY, JR, and SSH scrutinized and corrected the manuscript to its submission-ready version. All authors read and approved the final version of the manuscript prior to submission.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The collection of sugar beet plant material and all experimental procedures were conducted at the research field of our institute with official permission from the institutional authorities, in accordance with institutional, national, and international guidelines and legislation, including compliance with the IUCN Policy Statement on Research Involving Species at Risk of Extinction. Not applicable.

Competing interests

The authors declare no competing interests.

Consent for publication

Not applicable.

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